Laboratory 5: Permanent Magnet Generator Systems

version 3.0

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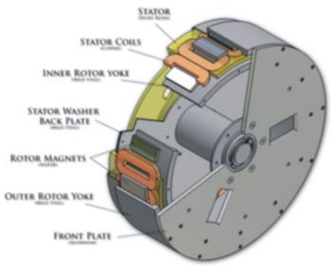
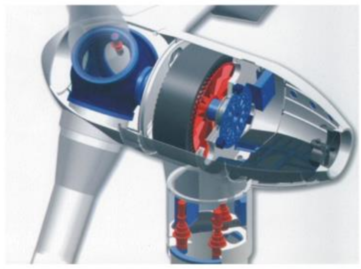
Introduction

Permanent Magnet (PM) Generators are used extensively in renewable energy applications from wind to hydro with sizes ranging from a few hundred watts to megawatts. In this laboratory session you will hopefully get an understanding of the basic operation of a PM Generator before investigating its application in stand-alone applications, examples being remote mountain bothy's, islands not connected to the national grid and off shore installations.





Aims and Objectives

Model 1: Stand Alone PM Generator + Diode Rectifier

- Relationship between pole number and output voltage
- Output voltage/frequency v generator speed
- Maximum power output v generator speed
- Diode Rectifier output voltage range v generator speed and load

Model 2: Variable Speed PM Generator Battery Charging System

- Wind turbine application – matching the power curve
- Controlling Buck Converter output voltage using PWM Duty Cycle

Model 1: Stand Alone PM Generator + Diode Rectifier

In this experiment we will investigate the operation of a variable speed 3-phase PM Generator connected to an uncontrolled diode rectifier to produce a variable DC voltage output from the variable frequency/variable voltage generator output

Variable Speed PM Generator + Diode Rectifier

We first of all want to investigate the basic relationship between generator speed, machine pole number and output voltage/frequency. A model (Lab5_Model1.bak) is available which implements this so load this onto your computer as before and open in Portunus. It should look like the circuit shown on Figure 1. The circuit consists of a 2 Pole PM generator (PSM1) connected to a 3-phase diode rectifier plus 100µF output capacitor (C1) to produce a smooth DC output voltage across an 8KΩ Resistive load (R1). The high load resistance results in extremely low output current from the generator hence the generator terminal voltage is approximately equal to the generator back EMF. This allows us to do no load tests on the generator. A speed source (NSRC1) drives the PM generator at a fixed speed.

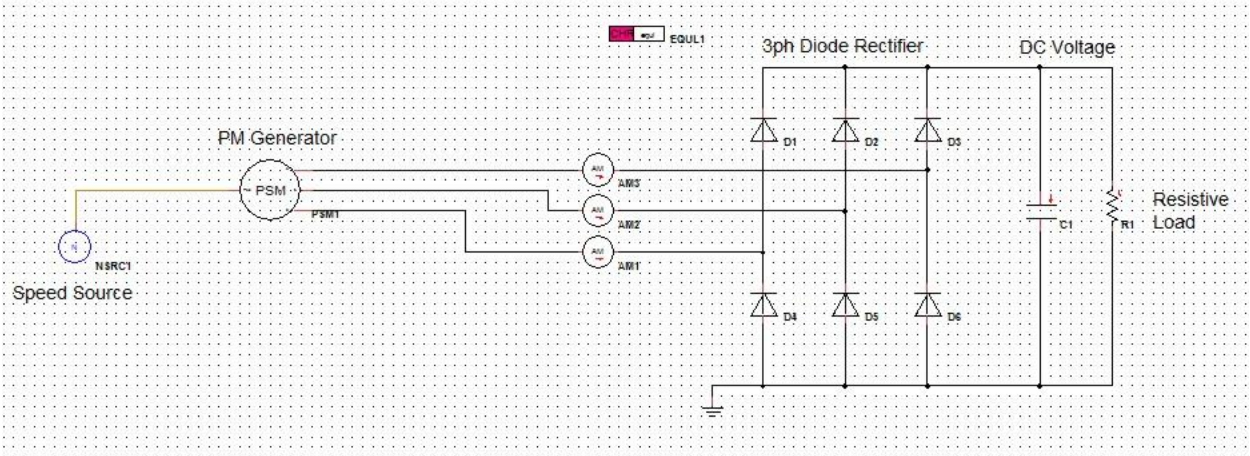


Figure 1: PM Generator + Diode Rectifier

Exercise 1.1: Output Voltage vs Generator Pole Number

We want to first of all determine the dc output voltage when operating at typical speeds found in small wind turbines (<500rpm). Set the speed source (NSRC1) to 400rpm.

Set **TEND = 5s** in the Simulation Control Panel [F9] and output the voltage across the load resistor (R1.V) on an On Sheet Display. Run the simulation to determine the output (DC) voltage across R1:

2 Pole PMG	Output Voltage (R1.V)	70.5375

Given that generator output power is proportional to $V_{ph}.I_{ph}$ and winding losses are proportional to I_{ph}^2 it is clearly beneficial to have as high an output voltage as possible from the generator for a given speed. How can we do this? Answer, increase the number of poles on the generator. Edit the PSM1 Parameter Panel to set Pole Pairs = 6 and rerun the simulation (set **TEND = 1.5s**).

12 Pole PMG	Output Voltage (R1.V)	425.712

Comment:

For the same speed of 400 rpm, increasing the generator from 2 poles to 12 poles raises the dc output voltage from about 70.5 V to 425.7 V, i.e. roughly in proportion to pole number. This higher voltage for a given speed allows the same power to be delivered at lower current (lower copper and rectifier losses), so small low-speed PM generators are normally designed with many poles.

Given the obvious advantage of the higher pole number machine it is no surprise to find that small PM generators will typically have >10 poles.

Exercise 1.2: Output Frequency vs Generator Speed.

We now want to validate a couple of basic equations for the PM generator connected to a 3-phase diode rectifier:

Generator Electrical Output Frequency:

$$Speed(rpm) = \frac{f_s.120}{P}$$

where:

Speed(rpm) = generator speed in rpm
f_s = electrical (voltage/current) output frequency in Hz
P = generator pole number (12 in this example)

Connect a Voltmeter (VM1) across any two phases of the generator output (see Figure 2) and display VM1.V on an On Sheet Display to measure the period of the output voltage. Set the generator speed using NSRC1 speed parameter and simulate at the following speeds. Using measured period calculate the frequency (Hz) of the generator output and compare with theoretical value given by the above equation:

Speed (rpm)	Output Period (ms) measured	Output Frequency (Hz) measured	Output Frequency (Hz) theory
100	100.0	10.0	10.0
200	50.0	20.0	20.0
400	25.0	40.0	40.0

Exercise 1.3: NO LOAD Rectifier Output Voltage vs Generator Speed

3 Phase Diode Rectifier (+ capacitor) DC Output Voltage:

$$V_{dc} = \sqrt{2}.V_{Lrms}$$

where:

V_{dc} = DC Output Voltage
V_{Lrms} = generator output rms line voltage

Use a Signal Analyser block (Sana1) to determine the rms of the generator line voltage (VM1.V) as shown on Figure 2. Note that the Sana block should be set to Sliding Window with Window Length equal to the corresponding period of the electrical output (voltage) for each speed.

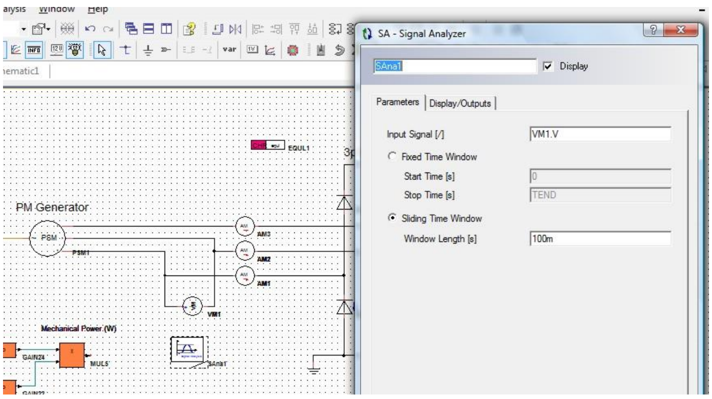


Figure 2: Measuring Generator rms Line Voltage

Output the instantaneous line voltage VM1.V and the rms line voltage onto an On sheet display and set **TEND = 1s.** Run the simulation at the following generator speeds remembering to change the Sliding Window Length each time to equal the line voltage period at each speed (determined in previous exercise).

Speed	Line Voltage Period	rms Line Voltage measured	Output DC Voltage (measured)	Output DC Voltage (theory)
(rpm)	(ms)	(V)	(V)	(V)
100	100.0	76.9465	106.124	108.818
200	50.0	153.887	213.075	217.629
400	25.0	307.758	425.712	435.235

Plot Output DC Voltage (measured and theoretical) v Speed (x axis) on the same graph, and plot Line Voltage Frequency (theoretical and measured) v Speed (x axis) on a separate graph. Comment on how the measured values compare with theory.

The measured DC output voltage and line voltage frequency both increase almost linearly with speed and are very close to the theoretical curves. The measured DC voltages are slightly lower than theory, which is likely due to diode drops, winding resistance and measurement or modelling errors, but overall the agreement is good.

Exercise 1.4 Maximum Generator Output Power vs Generator Speed

The output power dissipated in the load resistor (R1) connected to the output of the diode rectifier is equal to:

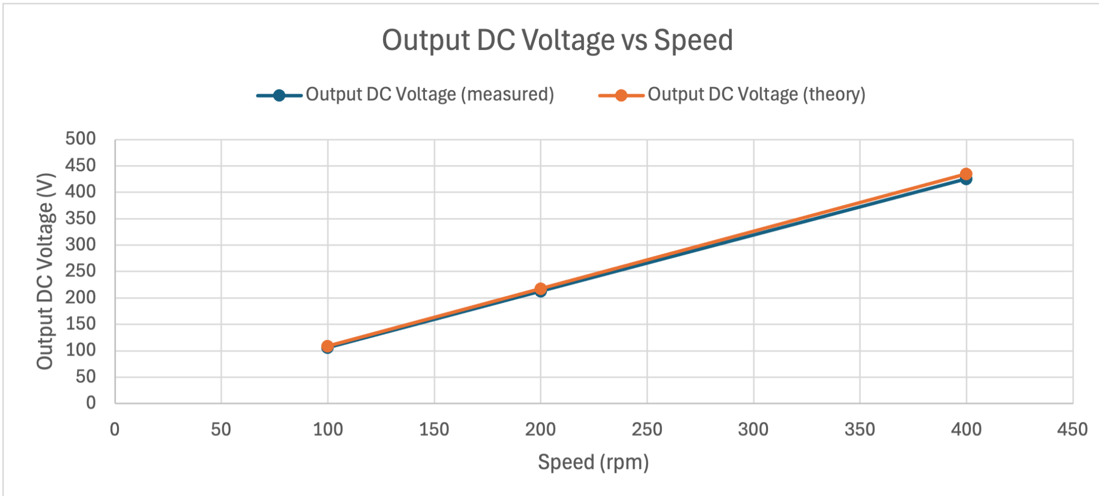
$$P_{load} = \frac{V_{dc}^{\ 2}}{R_{load}}$$

where:

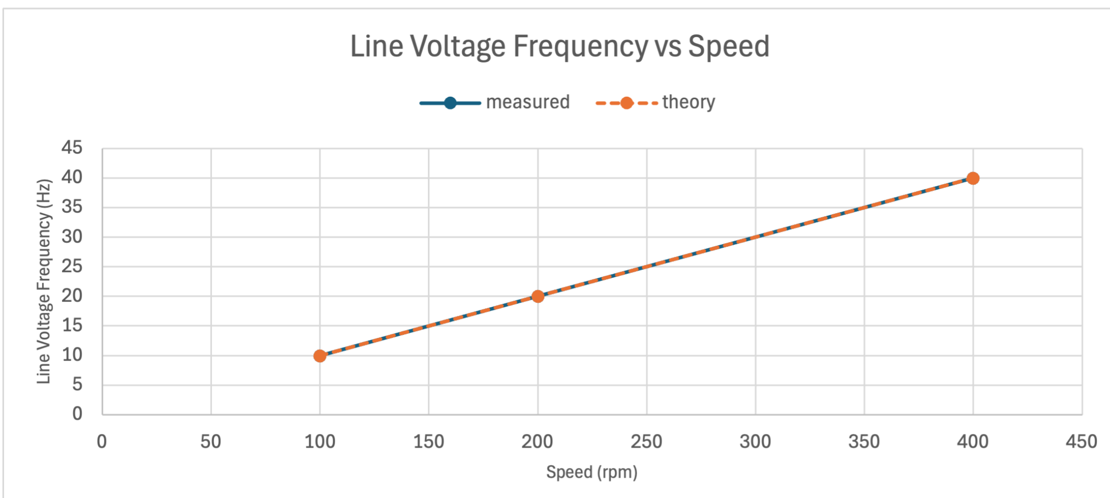
V_{dc} = R1.V
R_{load} = R1.R

This power is equal to the output (electrical) power of the generator minus any losses (diode rectifier losses in this case). Therefore, by changing the value of load resistance we can change the generator output power. What we find is that at any given generator speed there is a value of load resistance which results in maximum output power. To determine this maximum output power at a given generator speed we could change the load resistance by trial and error until the optimum value was found. An alternative (and more professional!) approach using Portunus is to 'sweep' the value of load resistance over a given time and setup Portunus to calculate the resultant load power. The resistance at which this maximum load power occurred can then be determined from the resultant simulation graphs.

The Load Resistance R1 value can be 'swept' through a particular range by applying a Sawtooth time signal (SAW1.OUT) to the R1 **Resistance** Parameter. Figure 3 outlines how to sweep the resistance from an initial value of 20.2Ω down to 0.2Ω over a 5 sec period using the Saw-Tooth Time Function (SAW1).



Output DC Voltage vs Speed
Output DC Voltage (measured)
Output DC Voltage (theory)
Output DC Voltage (V)
500
450
400
350
300
250
200
150
100
50
0
0
50
100
150
200
250
300
350
400
450
Speed (rpm)



Line Voltage Frequency vs Speed
measured
theory
Line Voltage Frequency (Hz)
45
40
35
30
25
20
15
10
5
0
0
50
100
150
200
250
300
350
400
450
Speed (rpm)

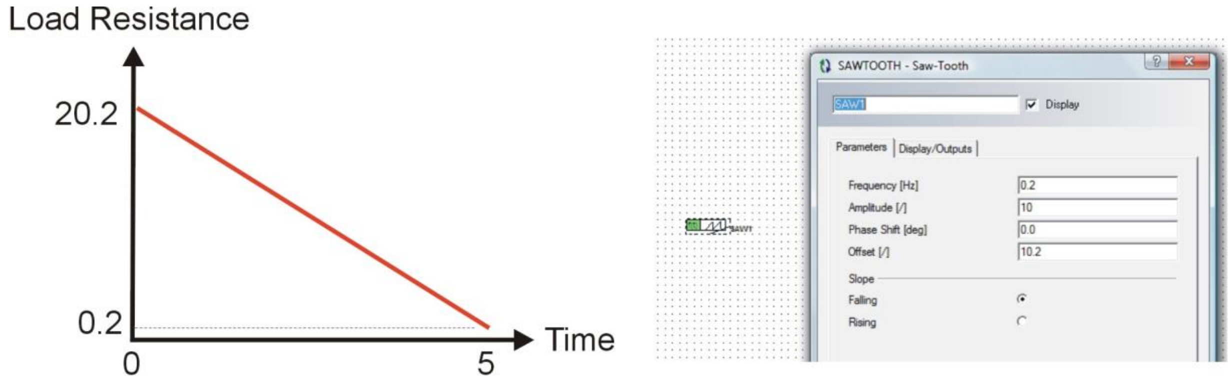
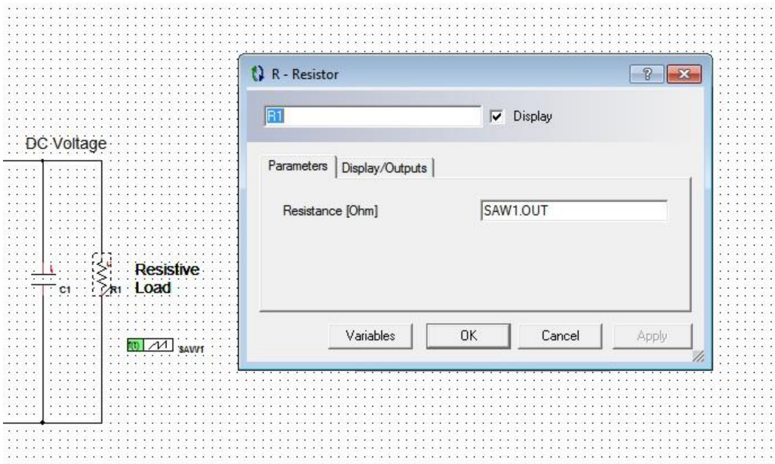


Figure 3: Swept Load Resistance

Sawtooth Parameters:

Parameter	Value
Frequency	0.2
Amplitude	10
Phase Shift	0
Offset	10.2
Slope	Falling

The Sawtooth output (SAW1.OUT) is allocated to R1.R as follows:



We also need to measure the average output power in Portunus. This can be done in a number of ways but for this I would suggest using the V^2/R equation shown earlier (but using the **Average** of the DC Link voltage as there is some ripple on it) as shown on Figure 4. Note that the **sliding window length** must be set for each generator speed as before.

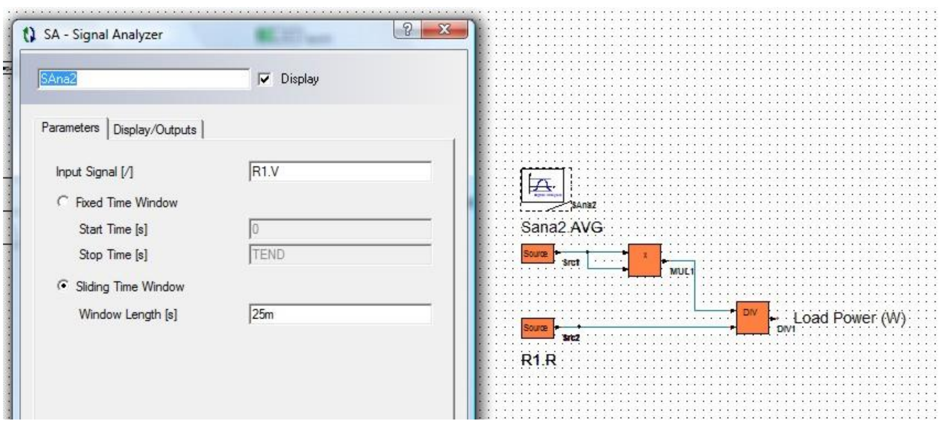


Figure 4: Measuring Load Power

Display Load/Output Power (DIV1.OUT), DC Output Voltage (R1.V) and the Load Resistance (SAW1.OUT) onto **separate** On Sheet Displays.

Change the Rectifier output Capacitor to 1000µF (the higher the value of Capacitance the lower the ripple voltage for a given load) and Set **TEND = 5s** in the Simulation Panel. To determine maximum power and the corresponding resistance value I suggest the following procedure at **EACH** speed:

1. Simulate over the 5s period and from the Output Power curve determine the time (T_{Pmax}) at which **maximum** power occurs.
2. Set the simulation time **TEND to T_{Pmax}** and re-simulate.
3. Use the Table option on Load Resistance (SAW1.OUT) On Sheet Display to determine the load resistance and DC Output Voltage at T_{Pmax} (ie final value)

Generator Speed	Maximum Power	DC Output Voltage	Load Resistance
(rpm)	(kW)	(V)	(Ω)
100	1.67	49.87	1.685
200	3.82	110.5	3.63
400	8.15	251.6	7.48

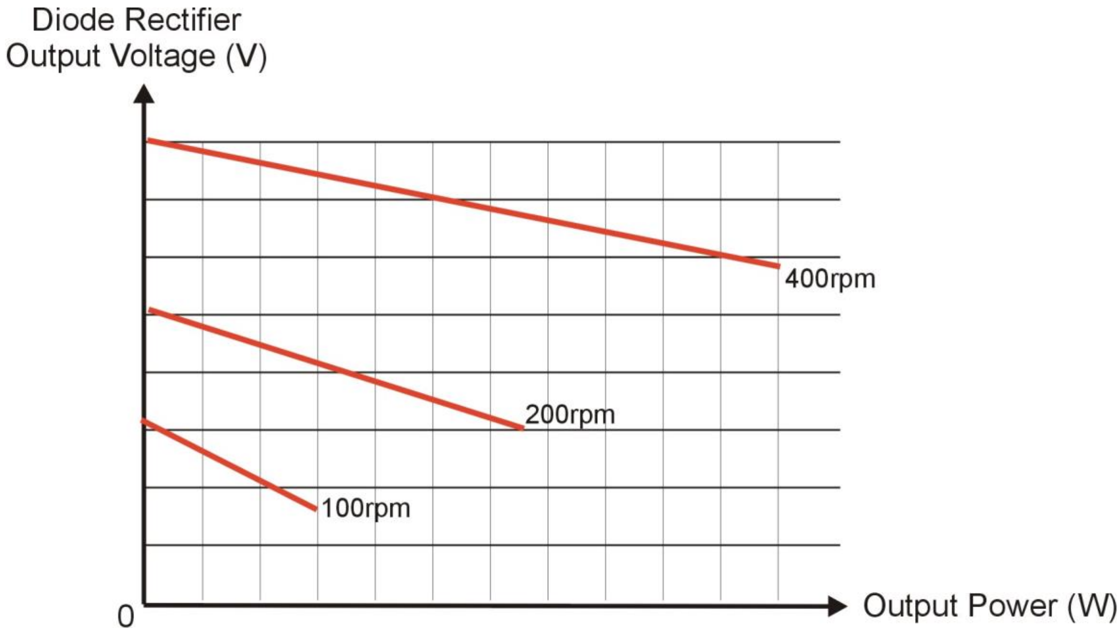
Rectifier Output (DC) Voltage Range

Given the results from the simulations so far fill out the following table and from this determine the Rectifier Output voltage range for operation between 100rpm and 400rpm.

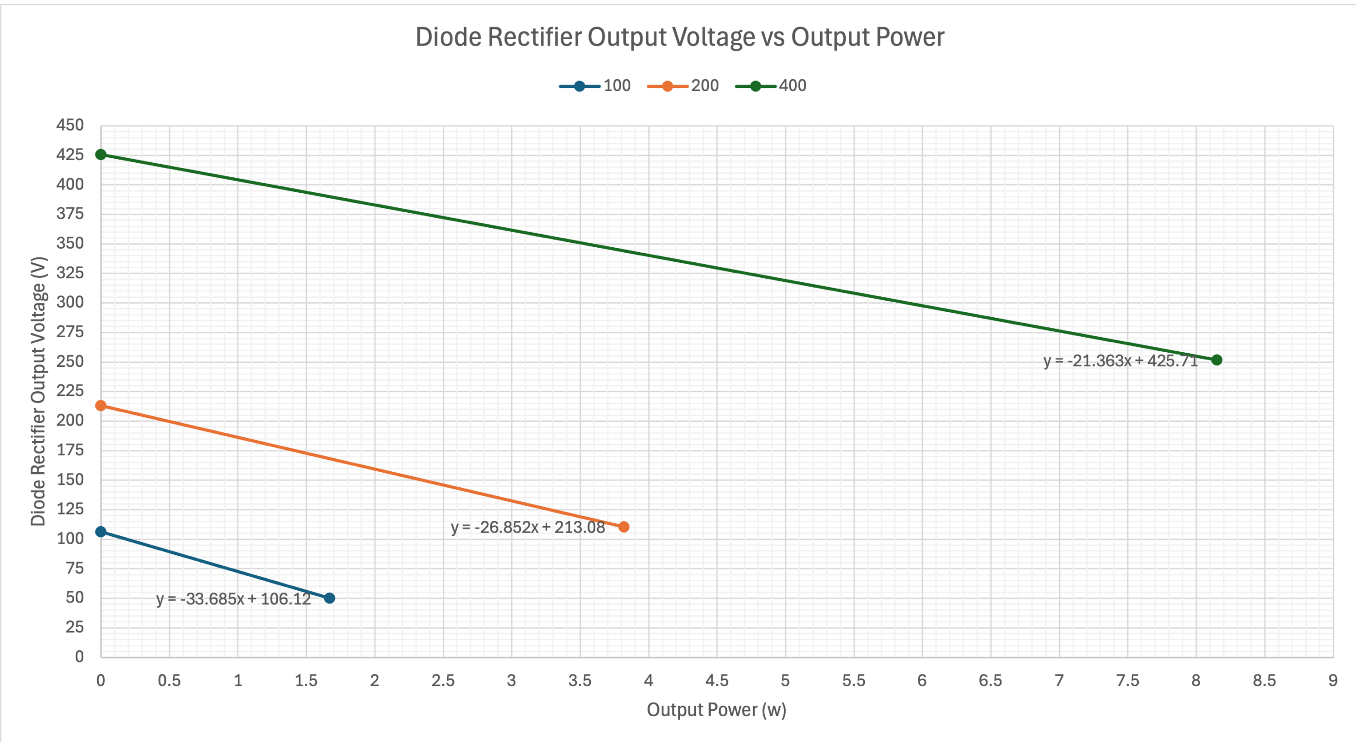
Generator Speed	Output Voltage @ No Load (Ex 1.3)	Output Voltage @ Max Power
(rpm)	(V)	(V)
100	106.124	49.87
200	213.075	110.5
400	425.712	251.6

Minimum DC Link Voltage (V)	49.87
Maximum DC Link Voltage (V)	425.712

Plots graphs of Diode Rectifier Output Voltage v Output Power for the three speeds using linear interpolation between no load and full load voltages (see Graph 1 as an example).



Graph 1: Rectifier Output (DC) Voltage v Output Power



Model 2: Variable Speed PM Wind Generator Battery Charging System

It is proposed that the 12-pole generator is used in a remote wind turbine application to provide battery charging capability. The complete system is shown on figure 5.

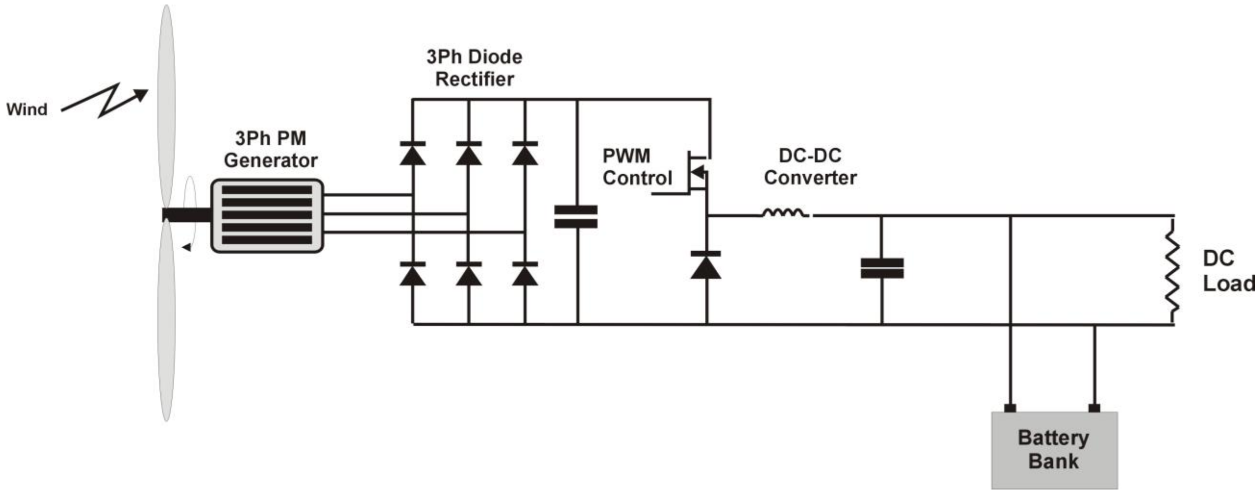


Figure 5: Stand Alone PM Wind Generator/Battery Charger

The system includes the existing PM generator and associated 3-phase diode rectifier. In addition, a DC-DC converter is required to interface to the battery bank, the requirements being to produce a fixed DC voltage (52V) to charge the 48V batteries and provide direct power (DC) to various DC loads (lighting and heating). Your results so far hopefully show that the diode rectifier outputs a range of voltage over the operating range of the generator therefore the additional DC-DC power stage is required to convert the variable DC Voltage to a fixed DC Voltage. Given that in this application the input voltage will always be higher than the required output voltage over the required operating range a Buck Converter is chosen. The topology for a Buck converter is shown on Figure 6 with associated component parameters given in Table 1.

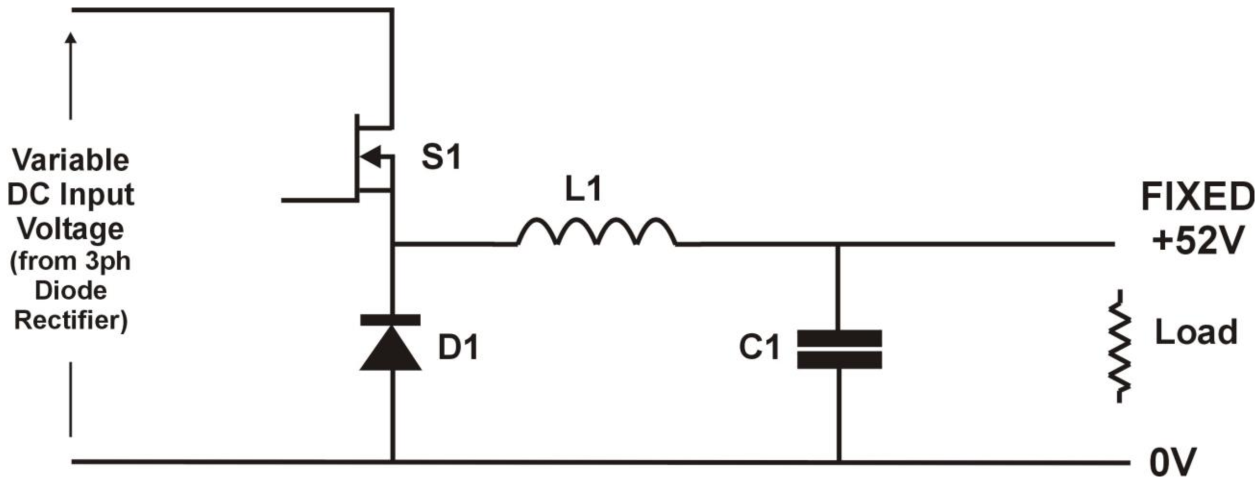


Figure 6: Buck DC-DC Converter

Component	**Parameter**	**Value**
S1 – Ideal Switch	Control Signal	PWM1.OUT
D1 – Diode	Characteristic	EQUL1
L1 – Inductor	Inductance (H)	500μ
C1 – Capacitor	Capacitance (F)	10000μ
PWM1 – PWM Time Function	Period Time (s) Duty	50μ 0.25 (this will change)

Table 1: Component Parameters

Add a model for the Buck converter to the output of the Diode Rectifier in Portunus, modifying all component parameters as shown. **Also remove the load resistance at the output of the Diode Rectifier and set the Rectifier output capacitance equal to 10000uF**.

I'm assuming that you have come across the Buck Converter in previous Courses, basically the output voltage is related to the input voltage and the Duty Cycle of the switch as follows:

$$V_{out} = D.V_{in}$$

where:

V_{out} = Output Voltage (V)
D = Duty Cycle of Switch
V_{in} = Input Voltage (V)

(Note that this equation is only valid in the Continuous Conduction mode of operation, but this will always be the case for this application).

Exercise 2.1: Constant Output Voltage operation of Buck Converter

The optimum operating curve for the Wind Turbine is shown on Figure 7.

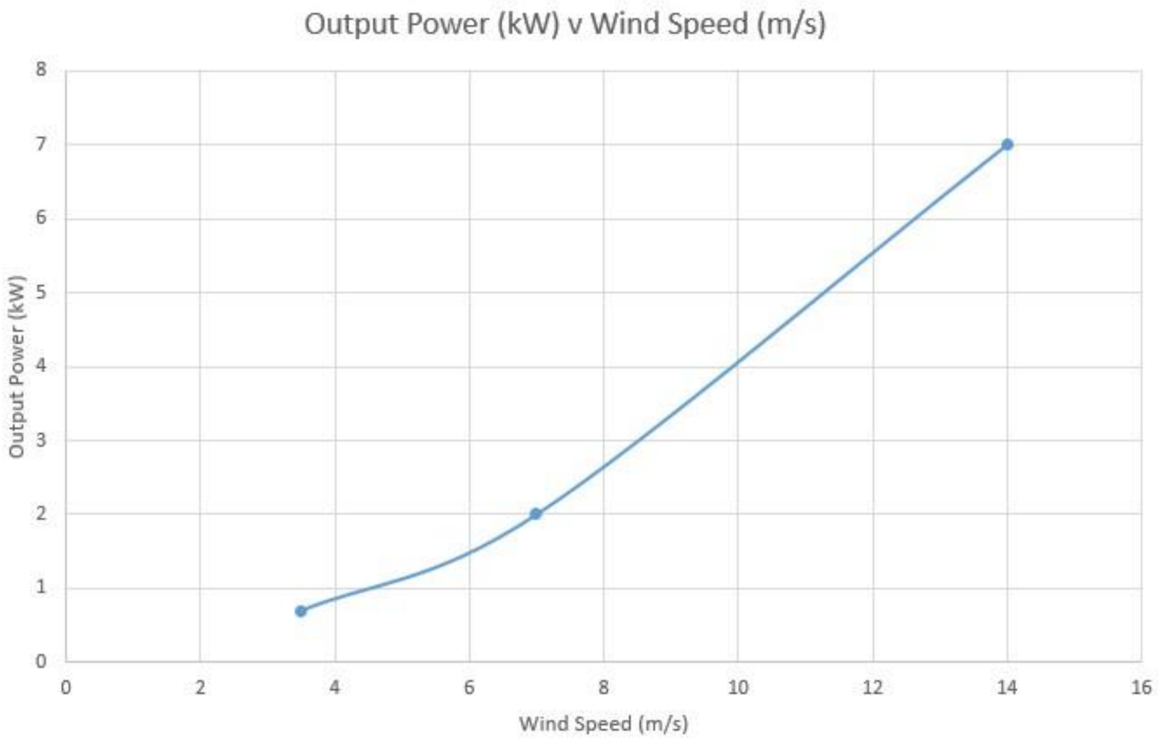


Figure 7: PM Wind Turbine Power Curve

Your challenge is to determine the necessary output load resistance and duty cycle combination to achieve 52V Output Voltage at the following points:

Wind Speed (m/s)	Turbine Speed (rpm)	Output Power (kW)
3.5	100	0.75
7.0	200	2.0
14.0	400	7.0

Procedure:

1. Calculate necessary resistance to achieve required output power @52V
2. Use the previous graphs of Diode Rectifier Output Voltage v Output Power to determine INPUT voltage to Buck Converter at this load power and speed (assumes Buck converter is 100% efficient).
3. Use the Buck Converter Voltage ratio equation to determine the required Duty Cycle.

Generator Speed	Output Power	Load Resistance	Rectifier Output Voltage	Duty
(rpm)	(kW)	Ω	(V)	
100	0.75	3.605	80.856	64.31%
200	2.0	1.352	159.376	32.63%
400	7.0	0.386	276.169	18.83%

Add a (Load) Resistor (resistance value set according to the calculations above) to the output of the Buck Converter and also add blocks to measure the output power of the Buck Converter using a similar technique to that used to measure the output power of the Diode Bridge (but without the need to average the Output Voltage as it is constant DC).

Display the following signals to separate On Sheet Displays:

- Buck Converter Input Voltage
- Buck Converter Output Voltage
- Buck Converter Output Power

Now, using the Portunus model, determine actual circuit performance at the given speeds using these values of load resistance and Duty.

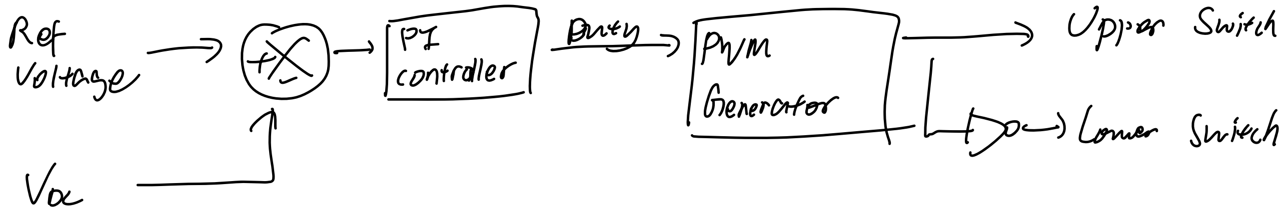
Set TEND = 0.3s

Generator Speed	Output Power	Load Resistance	Duty	Buck Input Voltage	Buck Output Voltage	Buck Output Power
(rpm)	(kW)	Ω		(V)	(V)	(kW)
100	0.75	3.605	64.31%	79.2356	56.21	0.863
200	2.0	1.352	32.63%	171.426	57.26	2.412
400	7.0	0.386	18.83%	252.752	55.79	7.948

How do the simulation results compare with the desired values for output voltage and power? What would you change in the simulations to improve agreement?

The simulated buck output voltages are close to the desired ~56–60 V but slightly higher, so the corresponding output powers are also above the targets (0.75, 2 and 7 kW). This is partially due to using approximate duty ratios, so the converter delivers a bit too much power. To improve agreement I would refine the duty cycle for each speed and include realistic losses (switch, diode and winding resistance) in the simulation.

Clearly some form of closed loop control is required to maintain a constant output voltage from the Buck Converter given changes to input voltage and load. Suggest (sketch) a suitable controller topology to achieve this.



Accessing Existing Models

In this laboratory session you are required to access an existing Portunus model (Lab5_Model1.bak) from Moodle page ENG4187. Copy these files into a known directory on your PC. To open this file in Portunus select 'File-Open' and go to the directory where you have stored the files. Note that in Portunus you should select 'all files' in the **Open File** panel to access the .bak file. If you subsequently want to save models then choose the Lab5_Model1**.bak** file as it is MUCH smaller than the corresponding .ecd file but contain all the necessary information and can be subsequently opened from Portunus using the select 'all files' in the **Open File** panel.